

HOW TO INTEGRATE BY PARTS IN 4 STEPS



ILATE

- I** Inverse Trigonometric
- L** Logarithmic
- A** Algebraic
- T** Simple Trigonometric
- E** Exponential

$$\int u \frac{dv}{dx} dx = uv - \int v \frac{du}{dx} dx \quad \text{or} \quad \int u dv = uv - \int v du$$

AAHL



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AHL 5.16

Content	Guidance, clarification and syllabus links
1. Integration by substitution.	On examination papers, substitutions will be provided if the integral is not of the form $\int kg'(x)f(g(x))dx$. Link to: integration by substitution (SL5.10).
2. ✓ Integration by parts.	Examples: $\int x \sin x dx$, $\int \ln x dx$, $\int \arcsin x dx$
Repeated integration by parts.	Examples: $\int x^2 e^x dx$ and $\int e^x \sin x dx$.



order of functions

ILATE

I**Inverse Trigonometric** $\sin^{-1} x, \cos^{-1} x, \tan^{-1} x \dots$ **L****Logarithmic** $\ln |x|, \ln |x^2 + 3x|, \log_a x$ **A****Algebraic** $x, x^2 + 5, x^3 + 3x^2 + x - 7$ **T****Simple Trigonometric** $\sin x, \cos x, \sec^2 x \dots$ **E****Exponential** $e^x, a^x, 3^x \dots$

$$\int \underset{(1)}{u} \underset{(2)}{\frac{dv}{dx}} dx = \underset{(1)}{uv} - \int \underset{(2)}{v} \underset{(1)}{\frac{du}{dx}} dx \quad \longrightarrow \quad v = \int \underset{(2)}{\frac{dv}{dx}} dx$$



Inverse Trigonometric



May be given as a single function



Logarithmic



May be given as a single function



$$e.g. \int \sin^{-1} x dx = \int \sin^{-1} x \cdot 1 dx$$



$$e.g. \int \ln |x| dx = \int \ln |x| \cdot 1 dx$$



Question (1):

[Maximum mark: 5]

Using integration by parts find $\int x^2 \cos x \, dx$.

$$\stackrel{(1)}{=} x^2 (\sin x) - \stackrel{(2)}{\int} (\sin x) (2x) \, dx$$

$$= x^2 \sin x - 2 \stackrel{(1)}{\int} \stackrel{(2)}{x \sin x} \, dx$$

$$= x^2 \sin x - 2 \left[x (-\cos x) - \int 1 \cdot (-\cos x) \, dx \right]$$

$$= x^2 \sin x + 2x \cos x - 2 \int \cos x \, dx$$

$$= x^2 \sin x + 2x \cos x - 2 \sin x + C$$

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$$\int u \frac{dv}{dx} \, dx = uv - \int v \frac{du}{dx} \, dx$$



Question (2):

[Maximum mark: 5]

Find $\int \arccos x \, dx$.

$$= \int \cos^{-1} x \cdot 1 \, dx$$

(1) (2)

$$= (\cos^{-1} x) \cdot (x) - \int (x) \cdot \left(-\frac{1}{\sqrt{1-x^2}} \right) dx$$

$$= x \cos^{-1} x - \frac{1}{2} \int \frac{-2x}{\sqrt{1-x^2}} dx$$

$$= x \cos^{-1} x - \frac{1}{2} \left(\cancel{2} \sqrt{1-x^2} \right) + c$$

$$= x \cos^{-1} x - \sqrt{1-x^2} + C$$

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$$\int u \frac{dv}{dx} dx = uv - \int v \frac{du}{dx} dx$$

$$\frac{d}{dx} (\cos^{-1} x) = \frac{-1}{\sqrt{1-x^2}}$$

$$u = 1 - x^2$$

$$\therefore \frac{du}{dx} = -2x$$

$$\int \frac{du}{\sqrt{u}} = \int u^{-1/2} du$$

$$= \frac{u^{1/2}}{1/2} + c$$

$$= 2\sqrt{u} + c$$



Question (3):

$$\int \ln |x| dx = \int \ln |x| \cdot \underset{(1)}{1} \underset{(2)}{dx}$$

$$= \ln |x| \cdot (x) - \int \cancel{(x)} \frac{1}{\cancel{x}} dx$$

$$= \boxed{x \ln |x| - x + c}$$

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$$\int u \frac{dv}{dx} dx = uv - \int v \frac{du}{dx} dx$$



Question (4):

[Maximum Marks: 5]

Evaluate $\int x^2 e^x dx$
(1) (2)

$$= x^2 (e^x) - \int (2x) (e^x) dx$$

$$= x^2 e^x - 2 \int x e^x dx$$

$$= x^2 e^x - 2 \left[x \cdot e^x - \int 1 \cdot e^x dx \right]$$

$$= x^2 e^x - 2x e^x + 2e^x + c$$

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$$\int u \frac{dv}{dx} dx = uv - \int v \frac{du}{dx} dx$$



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